

Multimodal Fashion & Context Retrieval — Technical Report

Assignment submission PDF source. Maps to deliverables: (1) Approaches, (2) Chosen architecture, (3) Codebase link, (4) Future work. Also covers modularity, scalability to 1M images, and zero-shot behavior.

Codebase (GitHub)

- Repository: <https://github.com/snigenigmatic/glanceML>
- Feature branch with this system: `cursor/fashion-retrieval-modal-5bcc`
- Indexing pipeline: `src/indexer/ (+ modal_app.py::build_index)`
- Retrieval pipeline: `src/retriever/ (+ modal_app.py::query / Gradio demo)`
- Evaluation: `src/eval/evaluate.py`, labels in `configs/relevance_labels.json`
- Reproducibility: see `README.md` (`modal run modal_app.py --stage ...`)

1. Approaches considered

Approach	Idea	Strengths	Weaknesses	When it fits
A. Vanilla CLIP / SigLIP	Single global image-text embedding, ANN search	Simple, zero-shot, scalable	Weak compositionality; confuses attribute binding (“red shirt, blue pants”)	Broad web search, coarse filters
B. Fashion-tuned CLIP (FashionCLIP / FashionSigLIP)	Same interface, domain-adapted contrastive training	Better garment/color/material recall	Still one vector; scene + multi-garment binding limited	Product catalog search
C. Caption-then-BM25/text	VLM captions indexed as text	Interpretable	Loses visual nuance; brittle phrasing	Small corpora, explainability focus
D. Hybrid: fashion embeddings + structured metadata	Dense recall + attribute filters/scores	Fixes compositionality without training a new model	Needs a metadata extractor; fusion weights matter	Fashion + context queries (this assignment)
E. Heavy VLM rerank of top-k	Ask a large VLM to score each candidate	Strong reasoning	Expensive at query time; overkill for internship scope	High-precision production rerank stage

Tradeoff summary: pure dense models scale and are zero-shot, but fail the assignment's compositional/fashion cases. Full generative reranking is accurate but slow/costly. Hybrid retrieval is the best precision-cost point: keep ANN for scale, add cheap structured scores for binding.

2. Chosen architecture

```
Images → FashionSigLIP embeddings
  → Florence-2 captions / detections → {color, clothing, scene, style}
  → ChromaDB (vectors + metadata)

Query → attribute parse + FashionSigLIP text embed
  → ANN top-50
  → metadata overlap score
  → score = 0.7·dense + 0.3·metadata
  → lightweight composition rerank on top-20
  → top-k
```

Why these models

1. **FashionSigLIP (Marqo/marqo-fashionSigLIP)** — fine-tuned from SigLIP with generalized contrastive learning on fashion attributes (category, color, material, keywords). Empirically stronger than FashionCLIP / generic CLIP on fashion retrieval benchmarks, while remaining a drop-in embedding model.
2. **Florence-2-base** — lightweight VLM with reliable detailed captioning and open-vocabulary detection. We do **not** treat it as a chatbot; we use task tokens (<MORE_DETAILED_CAPTION>, <OPEN_VOCABULARY_DETECTION>) and parse attributes deterministically. That keeps indexing reproducible and avoids an extra LLM.
3. **ChromaDB** — easiest persistent vector DB with metadata. Sufficient for 10^3 – 10^5 images; HNSW cosine search. At ~1M images, swap the store for Qdrant/FAISS IVF-PQ without changing retrieval math.
4. **Hybrid fusion + composition bonus** — dense score finds semantic neighbors; metadata score enforces multi-attribute constraints; reranker boosts candidates that jointly satisfy ≥ 2 constrained axes (the failure mode of bag-of-embedding CLIP).

How fashion queries are handled

- **Attribute-specific** (“bright yellow raincoat”) — color + garment terms in metadata; FashionSigLIP already biased toward fashion lexicon.
- **Contextual/place** (“business attire in a modern office”) — Florence-2/scene tags + style inference (business → formal).
- **Compositional** (“red tie and white shirt”) — both colors and both garments must appear for a high metadata/rerank score; dense-only cannot guarantee binding.
- **Style inference** (“casual weekend city walk”) — style aliases + scene priors (weekend→casual, city→street).

Shortcomings (acknowledged)

- Florence-2 captions can miss rare garments; PEDES seed captions help on clothing language, COCO object tags help on scene.
- Fusion weights (0.7/0.3) are heuristic; should be tuned on a labeled relevance set.
- Attribute parser is keyword-based (explicit, debuggable) but not open-vocabulary for novel slang.
- Pair/composition rerank can over-penalize when metadata is noisy (see `hybrid_rerank` vs `hybrid` below).

3. Dataset

- Source: **mixed corpus** — 500 CUHK-PEDES (attribute-rich person captions) + 300 COCO person images with scene cues (park / street / home / office).
- **800 images**, resized to 512px with letterbox padding.
- Axes covered: clothing/color language from PEDES + Florence captions; environments from COCO object cues + VLM scene tags.
- Open-vocab detection dumps were removed — they tagged ~16 garments on every image and made metadata scoring meaningless.
- Artifacts: `metadata.json` (seed labels), `index_meta.json` (VLM-enriched), Chroma persistence on Modal volume `glance-fashion-data`.

3b. Evaluation (what is graded)

Primary metrics: **Precision@K / Recall@K / AP@K** against hand labels in `configs/relevance_labels.json` (mixed-corpus `pedes_*` / `coco_*` IDs).

Compared systems: 1. Vanilla CLIP baseline (`openai/clip-vit-base-patch32`) 2. FashionSigLIP dense-only 3. Hybrid (dense + metadata) 4. Hybrid + composition / pair rerank

Attribute Coverage@K is a **diagnostic only** (self-grading against system metadata). Do not use it as the main claim.

Measured results (800-image mixed corpus, hand labels)

System	P@1	P@5	R@5	AP@5
CLIP baseline	0.80	0.20	0.14	0.18
FashionSigLIP dense-only	0.40	0.44	0.48	0.42
Hybrid	0.80	0.48	0.45	0.54
Hybrid + rerank	0.80	0.44	0.26	0.44

Takeaways: - Hybrid beats vanilla CLIP on P@5 / R@5 / AP@5 (0.48 / 0.45 / 0.54 vs 0.20 / 0.14 / 0.18) — the assignment claim. - q4 (casual city walk) is where metadata fusion helps most: hybrid P@5=1.00 vs CLIP 0.20. - Compositional q5: hybrid/dense retrieve the true white-blouse+red-tie match (`coco_3008`) at rank 2 and ahead of hard negatives; CLIP misses it

in top-10. Pair rerank currently hurts this query. - q1 still under-recalls: no literal yellow raincoat; hi-vis yellow vests are the closest proxies.

4. Scalability to 1M images

Stage	800 imgs	1M imgs
Embedding	batch GPU jobs	sharded Modal/map jobs
Metadata	Florence-2 once offline	same, async workers
Store	Chroma HNSW	Qdrant / FAISS IVF-HNSW + metadata index
Query	top-50 then rerank-20	unchanged compute profile (ANN + tiny rerank)

Retrieval logic is already ANN-first; only the backend index class needs to change.

5. Zero-shot behavior

No task-specific training on the eval prompts. FashionSigLIP + Florence-2 generalize from pretraining; attribute parsing is lexicon-based, not overfit to the five graded queries.

6. Future work

a. Locations (cities/places) and weather

- Add geo/place tags (EXIF, filename, or a places CNN / geocell classifier) as metadata fields.
- Condition retrieval with weather tokens (rain → raincoat/umbrella boost) via a small side model or OpenWeather lookup at query time.
- Expand query parser with place names and weather lexicon; fuse as a fifth axis with a low weight so it doesn't dominate garment match.

b. Precision improvements

- Learn fusion weights on a small judged set (logistic regression on [dense, meta, bonus]).
- Region-level embeddings (crop garments with Florence/YOLO, embed parts) for true color-garment binding.
- Hard-negative mining / fashion-specific reranker (cross-encoder) on top-20 only.
- Human feedback loop in the Gradio demo to collect relevance for continual tuning.

7. Reproducibility

See README.md for Modal commands. Core modules: src/indexer/build_index.py (Part A), src/retriever/hybrid.py (Part B).